

**TECHNICAL CHANGE SUMMARY**

Website Enhancements & Content Architecture Update

Comprehensive record of all client requested content updates, OEM portfolio expansions, tagline refactoring, and UI layout refinements delivered across the Windleaf platform.

1 Executive Summary

This report documents the full scope of enhancements performed across all six core pages of the Windleaf web application. The core objectives achieved include:

- **Accurate Industry Positioning:** Expanded the OEM footprint to represent all 8 major global wind turbine manufacturers and formalized owner-side IPP credentials.
- **Partnership Modernization:** Replaced historical legacy references with our forward-looking European design collaboration narrative.
- **Core Service Definitions:** Overhauled the “What We Do” service pillars into 9 specialized blade disciplines, rigorous Quality Inspection & Assurance, and Technical Due Diligence (TDD).
- **Information Hierarchy & Taglines:** Completed a sitewide audit ensuring zero repetitive taglines (eyebrows) and harmonizing section titles with clean multi-line typography.

2 Major Content & Brand Positioning Updates

A. Complete OEM & Manufacturing Footprint

Standardized Windleaf's extensive OEM manufacturing and field engineering background across all relevant sections.

Previous Copy	Suzlon • Vestas • Nordex • Siemens • GE • Goldwind (incomplete and unstandardized)
Delivered Copy	Suzlon • Vestas • Nordex • Goldwind • GE • Siemens Gamesa • RE Technologies • Envision
Where Applied	Homepage Experience Card, Services Practice Section, and Core Advantage Cards.

B. European Design Collaboration Narrative

Replaced all legacy references to single Danish design entities with a broadened, strategic partnership statement.

Previous Copy	Apex Wind Denmark / Collaboration with Apex Wind Denmark
Delivered Copy	Collaboration with European designers / European Design Collaboration
Where Applied	Site content profiles, About page milestones, Services blade design text, and Interactive 3D Globe pinpoints.

C. IPP (Owner-Side) Experience Clarification

Formalized the owner-side asset perspective specifically detailing TotalEnergies' multi-energy global footprint.

Updated Statement: "TotalEnergies acts as a massive Independent Power Producer (IPP), though it is officially classified as a global integrated multi-energy company."

3 Homepage “What We Do” Service Pillars Overhaul

Restructured the three primary service cards to provide depth and clarity on Windleaf's specialist capabilities:

Card 01 • Blade Engineering & Consulting

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DISCIPLINES

Comprehensive 9-discipline engineering coverage:

1. Blade Design & Engineering Review
2. Manufacturing Engineering
3. Quality Engineering & Assurance
4. Blade Inspection & Condition Assessment
5. Structural & Bonding Assessment
6. Blade Repair Engineering
7. Failure Investigation & Root Cause Analysis
8. Technical Due Diligence & Life Extension
9. Performance & Digital Engineering

Card 02 • Quality Inspection & Assurance UPDATED

Full Lifecycle Quality Rigor:

Comprehensive control of materials, manufacturing processes, composite laminates, bonding, dimensions, visual condition, NDT, repairs, documentation, traceability, and final inspection to ensure every component consistently meets specified quality, safety, reliability, and customer requirements.

Card 03 • Technical Due Diligence & Advisory (TDD)

ASSET INVESTMENT

Technical Due Diligence (TDD) provides an independent assessment of wind turbine blades, covering design, condition, defects, repairs, structural integrity, performance, remaining life, and technical risks to support informed investment and asset-management decisions.

4 Technology, AI & Robotics Flow Alignment

Established the three-tier operational sequence uniting advanced hardware with independent blade expertise:

STEP 01

Technology captures.

Drones, Crawlers & Internal Robotics

STEP 02

Engineering interprets.

AI Analytics & Senior Specialist Review

STEP 03

Windleaf delivers.

Actionable Repair Scope & Risk Mitigation

Refinements Made:

- **Typography:** Fixed the headline structure so all 3 core statements display on distinct, bold lines across hero and preview sections.
- **Section Tagline De-duplication:** Replaced repetitive Capabilities eyebrow on the capabilities grid with **THREE CORE PILLARS**, directly reinforcing the tri-pillar concept without repeating the header.

5 Sitewide Tagline (Eyebrow) Harmonization Audit

Every single section across the entire web application was inspected to eliminate redundant phrasing between the eyebrow tag and heading (H1/H2). Below is the verified matrix:

PAGE / ROUTE	SECTION HEADING (H1 / H2)	OPTIMIZED EYEBROW (TAGLINE)	STATUS
Home (/)	What We Do	SPECIALIST CAPABILITIES	✓ Distinct
Home (/)	Why should a client trust Windleaf?	THE WINDLEAF ADVANTAGE	✓ Distinct
Home (/)	Ready to Protect Your Blade Assets?	NEXT STEPS	✓ Distinct
How We Work (/how-we-work)	Independent, disciplined, end-to-end...	OUR APPROACH	✓ Distinct
How We Work (/how-we-work)	How We Deliver Every Assignment	OUR PROCESS	✓ Distinct
How We Work (/how-we-work)	Flexible Ways to Engage	FLEXIBLE ENGAGEMENT	✓ Distinct
Services (/services)	Specialist blade engineering across every stage...	LIFECYCLE EXPERTISE	✓ Distinct
Services (/services)	Our Experience Across Practice Areas	FIELD EXPERIENCE	✓ Distinct
About (/about)	Independent Blade Engineering for Global Wind Energy	WHO WE ARE	✓ Distinct
About (/about)	Track Record Across the Lifecycle	INDUSTRY BACKGROUND	✓ Distinct
About (/about)	Our Guiding Principles	WHAT DRIVES US	✓ Distinct
Technology (/ai-automation-robotics)	Current & Developing Capabilities	THREE CORE PILLARS	✓ Distinct
Contact (/contact)	Start a Conversation with Our Blade Engineers	GET IN TOUCH	✓ Distinct

6 UI & Layout Refinements

- **Desktop Single-Line Containment:** Formatted *"Why should a client trust Windleaf?"* to display seamlessly on a single line on desktop and tablet screens, preserving visual prestige.
- **Component Flexibility:** Upgraded the core `SectionHeading` component to dynamically accept custom styling and line-break wrappers without breaking accessibility or semantic SEO.
- **Gallery Image Cache Optimization:** Replaced local gallery placeholder imagery with dedicated high-resolution blade manufacturing assets, bypassing Next.js internal image caching with optimized cache busters.

7 Developer Handoff Package

Ready for Immediate Integration:

A complete, lightweight ZIP archive containing all modified source files, components, and media assets has been prepared for the development team at:

C:\Users\admin\Downloads\windleaf-updated-source.zip (~6.4 MB)

